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CIA - M6

Q-1 $f(x) = (x-3)^2$, $w = 0.5$, $c_1 = c_2 = 1$, $r_1 = r_2 = 0.5$

particle 1: $x = 0$, $v = 0$
particle 2: $x = 6$, $v = 0$
For $T = 0$

fitness for $x_1(0) = [0-3]^2 = 9$
$x_2(0) = 6 = [6-3]^2 = 9$

Both the fitness level is same, we go with whose initial distance is less than $x = 0$

gbest = 0
At start, pbest = current position
pbest 1 = 0
pbest 2 = 6

Iteration 1 ($t = 0 \rightarrow 1$)

a) Update velocities
$v(t+1) = wv_1(t) + c_1 r_1 (pbest - x(t)) + c_2 r_2 (gb - x(t))$

particle 1 - $v(1) = 0.5(0) + (0.5)(0.-0) + 0.5(0-0)$
$= 0$

particle 2 $\Rightarrow v(1) = 0.5 v_2(0) + (6-6) + 0.5(0-6)$
$= 0 + 0 + (-3) = -3$

b) Update pos
$x(t+1) = x(t) + v(t+1)$
$x(1) = x(0) + v(1)$

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p1 $x_1 = 0+0 = 0$
p2 $x_2 = 6+(-3) = +3$

c) Fitness at new pos

p1 $= f(0) = (0-3)^2 = 9$
p2 $= f(3) = (3-3)^2 = 0$

d) Updated Pbest
p1 = old pbest at 0 ($f=9$), new ~~f~~ $=9$ ~~$x=0$ pbest $=0$~~
p2 = old pbest at 6 ($f=9$) new $f=0$, $x=3$, pbest $=3$

e) update gbest
Choose which has lowest fitness
$x=3$

Iteration 2
p1 = $x(1)=0$ $v(1)=0$, pbest $=0$
p2 = $x(1)=+3$ $v(1)=-3$, pbest $=3$

a) Update velocities
P1 : $v(2) = 0.5v_1(1) + 0.5(\text{pbest} - x_1(1)) + 0.5(\text{gbest} - x(1))$

$= 0.5 \times 0 + 0.5(0-0) + 0.5(3-0)$
$= 1.5$

$P_2 = v(2) = 0.5v_2(1) + 0.5(\text{pbest} - x_2(1)) + 0.5(\text{gbest} - x_2(1))$
$= 0.5(-3) + 0.5(3-3) + 0.5(3-3)$
$+0.5(3-3)$
$= -1.5$

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b Updated distance $= x(t+1) = x(t) + v(t+1)$
$x(2) = x(1) + v(2)$

particle 1 $= 0 + 1.5 = 1.5$
particle 2: $3 - 1.5 = 1.5$

c Fitness at $t = 2$

$f(1.5) = (1.5 - 3)^2 = (1.5)^2 = 2.25$

d Updated pbest

p1: old pbest at 0 has $f = 9$, new $= 2.25$
pbest 1 $= 1.5$

p2 = old pbest $3 + 0$, 2.25 at $x = 3$
pbest $= 3$

e gbest $f = 0$, $x = 3$

Q-2 Minimize $f(x, y) = x^2 + y^2$
particle 1 : $(2, 2)$, $x = 0$ $v = 0$, $f = 8$
particle 2: $(1, 3)$, $x = 0$ $v = 0$, $f = 10$
particle 3: $(4, 2)$, $x = 0$, $v = 0$, $f = 20$
$w = 0.4$, $c1 = c2 = 1$, $r1 = r2 = 0.5$

Initial pbest and gbest:
Initially each particles pbest is its starting posn

particle 1: pbest $= (2, 2)$
particle 2: pbest $= (1, 3)$
particle 3: pbest $= (4, 2)$

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gbest = whose fitness is the minimum
$f = 8$ is the minimum
gbest = $(2,2)$
Iteration 1 :
① Velocity Update

$$v(t+1) = w v_i(t) + c_1 r_1 (pbest - x_i(t)) + c_2 r_2 (gbest - x_i(t))$$

$$= 0.4 v_1(0) + 0.5 (pbest - x_i(t)) + c_2 v_2 (gbest - x_i(t))$$

particle 1: $0.4 \times 0 + 0.5((2,2)-(2,2)) + 0.5(2,2-2,2)$
$v(1) = (0,0)$

particle 2 = $0.4 \times 0 + 0.5((1,3)-(1,3)) + 0.5((2,2)-(1,3))$
$v_1 = 0 + 0 + 0.5(1,-1)$ ~~= 0.5(0.5~~

$= (0.5, 0.5)$

particle 3 $v_1 = 0.4 \times 0 + 0.5((4,2)-(4,2)) + 0.5((2,2)-(4,2))$
$= 0 + \cancel{0} \; 0 + 0.5(-2,0)$
$= (-1,0)$

② $x(t+1) = x(t) + v(t+1)$
$x(1) = x(0) + v(1)$

$x(1)_1 = (2,2) + (0,0)$
$x(1)_2 = x(0) + v(1)$

$x(1)_1 = (2,2) + (0,0) = (2,2)$
$x(1)_2 = (1,3) + (0.5, -0.5) \; (1.5, 2.5)$
$x(1)_3 = (4,2) + (-1,0) = (3,2)$

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(3) Fitness T = 1

$f(2,2) = 2^2 + 2^2 = 8$

$f(1.5, 2.5) = 8.5$

$f(3,2) = \cancel{8}\ 13$

(4) Updates pbest

p1 = old was 8 to x(2,2) ~~*~~ new is 9 [so 8] (2,2), pbest (2,2)

p2 = (1.5, 2.5) because (8.5 < 10)

p3 (3,2)

(5) update gbest = (2,2)

Iteration 2:

Velocities

$v_1(2) = 0.4\,[v_i[t] + 0.5\,(pbest_i - x_i(1))$ $+ 0.5\,(gbest - x_i(1))]$

pt 1 x(1) = (2,2), x4(0,0), pbest = 02,2)

p1: $v_1(2) = 0.4\,(0,0) + 0.5((2,2)-(2,2))$ $+0.5(2,2-2,2)$

$v_1(2) = (0,0)$

p2 = $0.4\,(0.5,-0.5) + 0.5(1.5, 2.5 - 1.5, 2.5)$ $+ 0.5((2,2) - (1.5, 2.5))$

$= (0.2, -0.2) + (0,0) + (0.25, 0.25)$

$= \cancel{0} \; 0.45, -0.45$

$p3 = 0.4(-1,0) + 0.5((3,2)-(3,2)) + 0.5((3,2)-(3,2))$

$= (-0.9, 0)$

(2) updated pos

$x(t+1) = x(t) + v(t+1)$

$x(1+1) = x(1) + v(1+1)$

$x(2) = x(1) + v(2)$

$P_1 = x_1(2) = (2,2) + (0,0) = (2,2)$

$P_2 = x_2(2) = (1.5, 2.5) + (0.45, -0.45) = (1.95, 2.05)$

$P_3 = x_3(3) = (3,2) + (-0.9, 0) = (2.1, 2)$

(3) Fitness $t = 2$

$P_1 = 8$

$P_2 = \cancel{8.8}\ 8.005$

$P_3 = 8.41$

(4) ~~Update~~ Updated pbest

$p_1 = (2,2)$

$p_2 = 8.005 < 8.5$, pbest $(1.95, 2.05)$

$p_3 = 8.41 < 13$, pbest $(2.1, 2)$

(5) gbest (min fitness so far from all particles)

$= (2,2)$ fitness $= 8$

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Q-3

1) Build a Tour using Nearest Neighbour starting at A

Distances (symmetric)

from A → B : 10, C : 15, D = 20, E = 10

from B : C : 35, D = 25, E = 15

from C : D = 30, E = 20

from D : E = 15

Step by step NN (start at A)

start A + A

~~Unv~~

Unvisited {B, C, D, E}

from A nearest cities : B = 10, E = 10, ~~E~~ tive

Lets break the tie by choosing that comes alphabetically

first → B.

① A → B (10), Route A - B
cost 10

From B, unvisited = {C, D, E}

B → C = 35, B → D = 25, B → E = 17, nearest is E

② B → E (17), Route A → B → E
Cost = 10 + 17 = 27

From E unvisited = {C, D}

E → C = 20, E → D = 15, ~~Route is~~ nearest is D

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③ ~~E → A E~~ D (15)
Route A → B → E → D
cost = 27 + 15 = 42

From D unvisited = {C}
must go to C : D → C = 30

④ D → C (30)
Route A → B → E → D → C
Cost so far : 42 + 30 = 72
return to start C → A 15

⑤ C → A (15)
full tour ∴ A → B → E → D → C → A
NN Tour length = 72 + 15 = 87

So Nearest Neighbour tour from A
A → B → E → D → C → A, total length = 87

2) Find the shortest possible tour
Now we want true optimal tour
We fix start at A (because its a cycle, starting point doesn't matter)

Check all possible way to arrange B, C, D, E around it (there are 4!/2 = 12 distinct)

Instead of listing all 24, we can reason by neighbour of A

Check Promising Tours.

① A → B → D → E → C → A

Cost, A → B: 10, B → D: 25, D → E: 15, E → C: 20,
C → A: 15

Total = 10 + 25 + 15 + 20 + 15 = 85

② NN Tour we found:
A → B → E → D → C
cost: 87

③ Other patterns all come out ≥ 95, 97, 100.
Smallest cost is 85
[A → B → D → E → C → A

Q4] Point (x, y)

H (HOME)	(0,0)
H_1	(3,4)
H_2	(10,0)
H_3	(5,12)
H_4	(0,8)

Distance between two points (x_a, y_a) (x_b, y_b)

$$d = \sqrt{(x_a - x_b)^2 + (y_a - y_b)^2}$$

Distance Matrix Lookup table

From \ To	H	H_1	H_2	H_3	H_4
H	—	5	10	13	8
H_1	5	-	7.81	8.06	5
H_2	10	7.81	~~8.06~~	13.93	12.81
H_3	13	8.06	8.06	—	5
H_4	8	5	5	5	—

∴ If we follow nearest neighbour heuristic Starting at 'H'

STEP	CURRENT	Unvisited	Nearest choice	Distance
1	H	H_1, H_2, H_3, H_4	H_1 [5]	5
2	H_1	H_2 H_3 H_4	H_4, 5	5
3	H_4	H_2, H_3	H_3, 5	5
4	H_3	H_2	H_2 [13.93]	13.93
5	H_2	-back to H	H =10	10

NN Tour :

$H \rightarrow H_1 - H_4 - H_3 - H_2 - H$

total distance = 38.93 mm